

GYEONGBIN RYOO

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Education

Dartmouth College, Hanover, NH

Master of Science, Computer Science (Expected June 2027)

- **Relevant Coursework:** AI Agents, Video Understanding, Visual Computing Seminar, Artificial Intelligence, Darkside of AI, Computer Vision, Computational Photography

Sungshin Women's University, Seoul, South Korea

Bachelor of Science, Artificial Intelligence Convergence (Graduated Feb 2025) — **GPA: 4.07/4.5**

- **Relevant Coursework:** Deep Learning, Computer Vision, Big Data Analysis, Natural Language Processing

Technical Skills

Programming: Python, C++, Java, JavaScript (Node.js, HTML/CSS)

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, OpenCV

Cloud & Tools: AWS (EC2, S3, Lambda), Git, Docker, SQL

Honors & Awards

- Top Academic Achievement Scholarship — Sungshin Women's University (90% tuition), 2024
- Undergraduate Excellent Paper Award (Encouragement) — KICS Summer Conference, 2024
- Academic Excellence I Scholarship — Sungshin Women's University (80% tuition), 2023
- Software Development Competition (Encouragement Award) — Dept. of AI Convergence, 2023

Technical Project Experience

Hyundai NGV (Affiliated with Hyundai Motors), Seoul, South Korea

Undergraduate Researcher (Industry-Academia Collaboration) | June 2024 – January 2025

- Enhanced **business models and integration scenarios** for robot-friendly buildings (delivery, parking)
- Contributed to technical standards, and 10 domestic patent applications jointly filed with Hyundai Motor Company (pending publication), along with simulations of robotic services

Ministry of Science and ICT, Seoul, South Korea

Undergraduate Researcher (Industry-Academia Collaboration) | March 2023 – February 2024

- Developed **AI algorithms for marine mobility solutions** to support waste collection and recycling in coastal waters.
- Analyzed ship movements as **time-series data** using Python, NumPy, Pandas, and PyTorch to optimize mobility designs.

Hyundai Heavy Machinery, Seoul, South Korea

Undergraduate Researcher (Industry-Academia Collaboration) | March 2023 – November 2023

- Developed **authentication and administrative modules** for a sloshing-load prediction system using **C# and AWS**.
- Built **ingestion and inference pipelines** to upload AI models and **visualize real-time predictions**.

Research & Teaching Experience

Teaching Assistant, Dartmouth College, New Hampshire, United States

Graduate Teaching Assistant for Artificial Intelligence | September 2025 – November 2025

Graduate Teaching Assistant for Advanced NLP | January 2026 – March 2026

Graduate Teaching Assistant for Machine Learning and Statistical Data Analysis | April 2026 – June 2026

Mobility Service Research Lab, Sungshin Women's University, Seoul, South Korea

Undergraduate Research Assistant | December 2023 – January 2025

- Researched effectiveness of **synthetic data** from **metaverse environments** to train AI models for real-world **autonomous driving**.
- Achieved **39% higher object detection** and **4% higher segmentation accuracy** via a hybrid data approach, leveraging **NVIDIA Isaac Sim** — a 3D simulation tool — to generate synthetic training data for autonomous driving AI models.
- Collaborated with the mobility engineering team to align AI model outputs with real-world autonomous driving scenarios
- Presented first-author research via poster presentations at a domestic and international conferences

Marine Hydrodynamics AI Research Lab, Sungshin Women's University, Seoul, South Korea

Undergraduate Research Assistant (Funded by UROP) | March 2023 – December 2023

- Predicted **sloshing impact loads** using regression to improve safety and reliability in marine environments
- Analyzed **6-DoF ship motions** with experimental data, preprocessed large-scale datasets, and applied AI models (**LightGBM, GAMs**) to boost prediction accuracy.
- Partnered with fellow data mining team members to refine regression models and validate prediction accuracy

Leadership & Activities

- Editor-in-Chief, Sungshin Women's University English Newspaper, Seoul, South Korea | December 2022 – February 2025
- Team Leader & Mentor, Naver Connect Foundation AI Camp, Yongin, South Korea | January 2023 – February 2023
- Python Instructor, Coding Education Volunteer Program, Seoul, South Korea | February 2022 – July 2022